Are We Chasing Ghosts? Quantifying Unattributable Polarization and Attributing the Rest to Annotator Groups

Anonymous ACL submission

Abstract

Polarization has recently been proposed as a robust way of distinguishing minor disagreements from systematic differences in opinion for imbalanced (minority) groups but there currently exists no methodology for attributing polarization to these groups. In this study, we discover and investigate two major limitations in realistic settings: (1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that attributes polarization to any annotator group while avoiding these limitations, and provide an open-source Python implementation. We apply our method to four subjective NLP datasets and confirm that gender and race consistently explain polarization patterns, while differences between annotator groups become stronger as the groups are further apart. Our results overall suggest a need for more annotations per item in subjective NLP datasets, although we empirically find that no more than 20 annotators per item are needed for reliable estimation.

1 Introduction

Annotations are essential for subjective tasks, such as content moderation, toxicity, and hate speech detection which are especially important for protecting minority groups (United Nations, 2025; Dioum, 2025). However, in practice analysis rests on inter-annotator agreement or majority-vote (Ivey et al., 2025; Giannoutsos et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" (Sap et al., 2019)), or discard *fundamental* disagreements between minority groups and the majority (Quaranto, 2022), (see the example in Fig. 1).

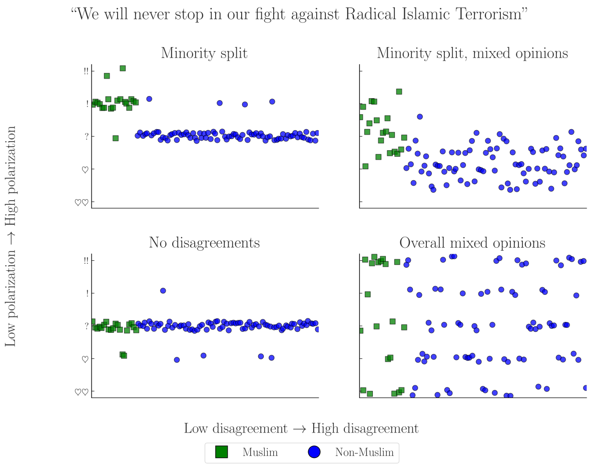

Figure 1: Simulated experiment exploring four scenarios of a hate speech detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question (RQ):

RQ. *How much polarization can we practically attribute to annotator subgroups given existing datasets?*

While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups our study discovers the existence of "inherent polarization", which cannot be attributed to *any possible group* of annotators–known or unknown (Finding 1). We also discover that polarization across comments has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2). These issues highlight that current subjective Natural Language Processing (NLP) datasets may lack the necessary number of annotators per comment–estimated empirically to be ≈ 20 annotators (§4.4).

In order to solve these issues we introduce a new metric—"*apunim*"—which attributes polarization to specific annotator groups. Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables both *quantitative* comparisons across datasets and statistical analysis via p-value estimation. Even in cases where unattributable polarization exists, *apunim* is capable of detecting general patterns of polarization across multiple groups.

We then apply our metric to four NLP datasets focused on Toxicity and Hate Speech detection (§5), finding that annotator gender and race significantly impact annotation (Finding 3), and that fundamental disagreements grow more pronounced the further apart annotator groups are (e.g., irreligious vs. very religious annotators–Finding 4). Finally, we release an open source Python (PyPi) library that implements our metric.[1] The code for the experiments, graph creation, and analysis can be found on our project's repository.[2]

Warning: This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.

2 Background and related work

2.1 Agreement

Popular inter-annotator agreement metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds, producing results that are difficult to compare across studies, being sensitive to class imbalance, and relying on questionable assumptions about chance agreement (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017; Wong et al., 2021).

Recent work has addressed some of these limitations through simpler statistical approaches (van der Velden et al., 2025), subgroup-specific adaptations of Cohen's kappa (Wong et al., 2021), and Bayesian models that explicitly separate subgroup disagreement from annotation noise (Ivey et al., 2025). While these methods improve applicability and robustness, they often come at the cost of interpretability.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still necessitates traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos and Likas, 2024). nDFU takes values in the range $[0, 1]$, where values close to 0 indicate little or no polarization (i.e., opinions are concentrated around a single mode), while values close to 1 indicate strong polarization with multiple distinct peaks. The metric is defined as follows:

[1] Library: `https://anonymous.4open.science/r/apunim-5F70`, Webpage: `https://anonymous.4open.science/r/apunim-page`

[2] Replication code: `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`

$$\mathrm{nDFU}(X) = \frac{\max\limits_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}$$
$$m = \arg\max_i f_i \tag{1}$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2 for an example).

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing polarization to subgroups is a relatively new idea. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are "causing" the exhibited polarization. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split. Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[3]

Another approach (Anonymous, 2026) attempts to first filter out non-substantial disagreements using polarization, and then analyzes differences between LLMs and annotator subgroups through traditional agreement metrics. This analysis however is limited by the inherent issues of using traditional agreement metrics for minority groups (§2.1), and the authors prioritize comparisons between human and LLM annotators instead of a deeper analysis of minority groups.

[3]For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to .001, and cases where attribution is .49.

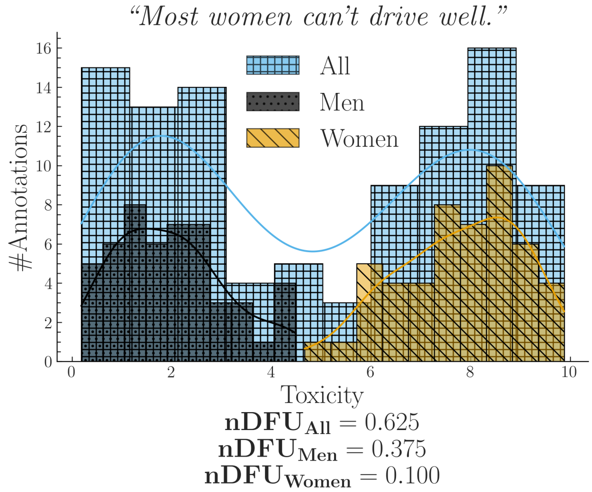

Figure 2: Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender.

3 Is it possible to attribute all polarization to annotator groups?

3.1 Problem Formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated comments.[4] Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5). As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{"Could be better, could be worse"}) \\ = \{ \ (+, F), (-, M), (+, F), \ldots \ \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c \mid g) = \{a(c, g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2.[5] The annotations

[4]These items could be of any modality, such as comments in a discussion, images, videos or survey questions, even though we consider comments in this study.

[5]All simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and std dev–see `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`.

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to sociodemographic dimensions (e.g., age, gender). Tasks are either toxicity (Tox) or hate speech detection (HS).

are generally polarized ($nDFU_{all} = 0.62$); but both female ($nDFU_{women} = 0.10$) and male ($nDFU_{men} = 0.37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annotators.[6]

3.2 Datasets

We use four datasets from current NLP research that include sociodemographic information at the level of individual annotators; specifically, the "Kumar" (Kumar et al., 2021) and "Sap" (Sap et al., 2022) datasets, which are concerned with Toxicity and hate speech detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.[7] While the toxicity and hate speech detection tasks are not equivalent, they share substantial overlap, as both require annotators to assess the presence and severity of harmful or abusive language.

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct sociodemographic dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (App. B; Fig. 7).[8]

3.3 Inherent polarization

3.3.1 What is it?

Inherent polarization refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of inherent polarization would invalidate this theory, since there would be no possible group or combination of groups g *s.t.* $nDFU(g) = 0$.

Definition 1: Inherent polarization. *Polarization that cannot be explained by any known or latent annotator grouping.*

Formally, the inherent polarization for a single comment c is given by:

$$Pol_{inh}(c) = \min_{i}\{nDFU(\tilde{r}_i(A(c)))\} \quad (2)$$

where $\tilde{r}$ is the random partition operator.

3.3.2 Experimental setup

We systematically explore the space of annotator groups by generating random partitions for each comment. When considered exhaustively, these theoretical, random groups include both observed attributes (e.g., men and women) and latent or unobserved factors (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators, this yields $\sum_{k=3}^{n}\binom{n}{k}$ possible groups, over which we test whether *any* exhibit intra-group polarization:

$$Pol_{inh}(c) = \min_{S \subseteq A(c),\, |S| \geq 3}\{nDFU(S)\} \quad (3)$$

Since enumerating all subsets is computationally intractable for large annotator counts,

[6]The continued polarization observed within the male annotators may suggest that a specific subgroup of men could also contribute to this polarization.

[7]For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to hate speech detection.

[8]This systematic discrepancy could be attributed to the DICES datasets being composed of LLM-generated dialogues with already-aligned models tuned to avoid unsafe speech. Even if such speech exists, it is likely much easier to distinguish than in human comments.

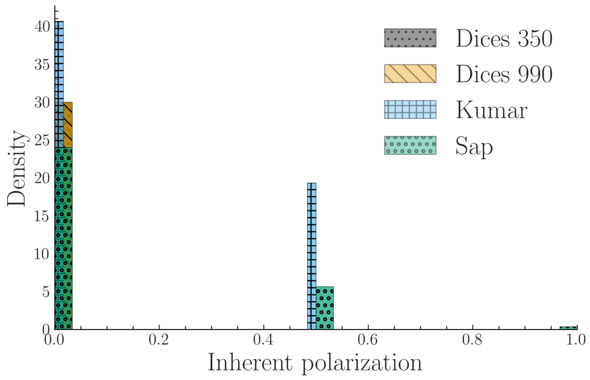

Figure 3: Inherent polarization across our datasets (§3.2). Contrary to assumptions made in prior work, inherent polarization is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

we approximate this quantity via Monte Carlo sampling when $|A(c)| \geq 10$ by drawing 1000 random partitions with random sizes:

$$Pol_{inh}^{MC}(c) = \min_{j=1}^{1000} \{nDFU(\tilde{r}_j(A(c)))\} \quad (4)$$

Because Eq. 4 considers only a strict subset of the total possible configurations, it provides a lower bound for Eq. 3:

$$Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \quad (5)$$

3.3.3 Does it exist?

We compute the inherent polarization for each comment across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 3), whereas for the two DICES datasets we rely on the Monte Carlo approximation (Eq. 4). Fig. 3 shows that in all DICES comments there exists at least one group that fully accounts for the observed polarization.[9] Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. Further experiments show that repeating the experiment for subsets of (5–50) annotators sampled ten times with replacement also lead to zero inherent polarization in the DICES datasets.

Finding 1. *Comments may exhibit a degree of inherent polarization that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

3.3.4 Why does it exist?

We hypothesize that this discrepancy is caused by the existence of a latent group which is not represented in the annotator population i.e., although a latent group could in principle explain the observed polarization, such a group may be *not represented when represented by only one or two annotators* (see App. C.2). This hypothesis explains why the Kumar and Sap datasets, which feature low annotator counts, contain comments exhibiting non-zero inherent polarization. In contrast, the DICES datasets are much less polarized (see App. B; Fig. 7 and Footnote 8); hence, all polarization can be explained even when a low annotator count is sampled for each comment.

This finding suggests that current NLP datasets may be *lacking fundamental information* for analyzing annotator behavior, especially in very subjective and sensitive tasks. Even though the Kumar dataset employs an impressive 17,280 annotators in total, the lack of annotations per comment results in a large section of these comments featuring unattributable polarization.

3.4 Polarization is directional

One way of potentially overcoming annotator sparsity, is aggregating annotations between comments. Fig. 4 shows an experiment where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might balance each other out, leading to a false negative. Therefore, we should never aggregate annotations across comments.[10]

Finding 2. *Polarization has a unique direction for each comment which may compromise cross-comment aggregation.*

4 How do we attribute the rest?

4.1 Apriori polarization

Instead of using inherent polarization, which represents the minimum possible polarization

[9] $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 5. The same logic applies if we tried to increase the number of random partitions in Eq. 4.

[10] A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

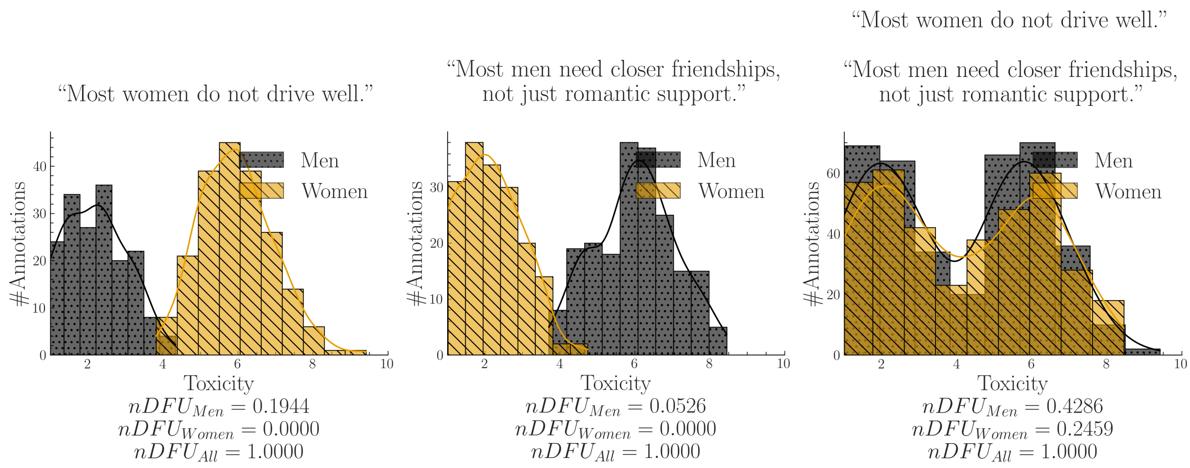

Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups.*

Definition 2: Apriori polarization. *The expected polarization of a comment's annotations.*

Specifically, we define apriori polarization for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \quad (6)$$

where $\tilde{r}$ is the random partition operator (similar to Eq. 2, but with matching group sizes). See App. C.1 for more details.

4.2 Expanding to the whole dataset

We can estimate how much polarization is generally present in the dataset by simply averaging the *apriori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(d) = \frac{1}{|d|}\sum_{c \in d} Pol_{apr}(c) \quad (7)$$

Which we will compare with the average observed polarization of all comments:

$$Pol_{obs}(d, g) = \frac{1}{|d|}\sum_{c \in d} nDFU(A(c \mid g)) \quad (8)$$

We can now compare the polarization that is attributed to a single annotator group (Equation 8) to a prior one (Equation 7). We define our normalized metric, *apunim*, as:

$$apunim(d, g) = \frac{Pol_{obs}(d, g) - Pol_{apr}(d)}{1 - Pol_{apr}(d)} \quad (9)$$

Definition 3: Apunim. *The degree to which an annotator group contributes to overall polarization in a set of comments.*

Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- $apunim(\theta) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(\theta) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(\theta) < 0$: This occurs when intra-group polarization is higher than apriori polarization; indicating that annotators of that group are polarized between themselves.

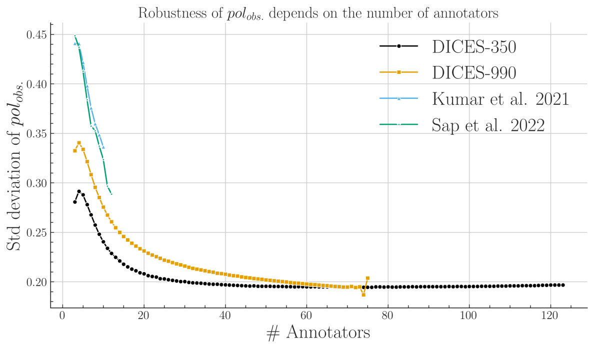

Figure 5: Standard deviation of 30 pol_{obs} values (Eq. 8). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (App. B).

4.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$H_0 : \overline{apunim}_{\text{rand}}(d, g) = apunim(d, g)$$

$$H_a : \overline{apunim}_{\text{rand}}(d, g) \neq apunim(d, g)$$

where $\overline{apunim}_{\text{rand}}(d, g)$ denotes the average apunim values obtained from t randomized annotator partitions. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.3.

4.4 How many annotators do we need?

Fig. 5 shows the impact of annotator count on the reliability of polarization estimation. We observe a clear, non-linear, inverse relationship between the number of annotators and the standard error of pol_{obs} across all datasets, which can be attributed to better histogram estimation employed by nDFU (§4.1) although the rate of improvement becomes increasingly marginal after 20 annotators. Interestingly, some variability in polarization persists even with an extraordinary amount of annotators, which we hypothesize to be caused by inherent noise in the dataset.

This finding reinforces the fact that most NLP datasets are lacking fine-grained information, as discussed in §3.3.3. However, the indication that only 20 annotators are needed for robust estimations of polarization is encouraging; it is much easier to scale up to this number of annotators compared to creating specializing datasets such as DICES which feature up to 100 annotators. Indeed, our formulation enables us to attribute some polarization even in cases with very low annotator counts, as shown in the next section.

5 Attributing real-life polarization to human groups

5.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided sociodemographic dimensions can partially explain polarization in the annotations.[11] Note that some groups are not included because there were no polarized items annotated by more than one annotator of these groups (see App. C.2). See App. D.3 for details on other parameters and execution times, and App. E for the full results.

5.2 Which factors contribute to polarization?

Ethnicity Across all datasets apart from Sap, we observe a consistent pattern in which annotation polarization is linked to annotator ethnicity, and particularly by AA annotators. In fact, *ethnicity is the strongest contributor to polarization* in the DICES datasets (AAs: $-.0428$/Asians: .0659 for DICES-350, all ethnicities for DICES-990) and the second strongest for Kumar (AAs: .2072). These results align with prior work suggesting that ethnic background is a systematic source of annotator disagreement in these tasks (Goyal et al., 2022).

Gender Men and women have statistically significant, opposite effects both in DICES-990 (Men: .028; Women: $-.025$) and Sap (Men: .091, Women: $-.244$), consistent with previous findings regarding hate speech (Wojatzki et al., 2018) and toxicity annotation (Binns et al., 2017). *Transgender* anno-

[11] We sample 60,000 comments from the Kumar dataset due to memory constraints.

Group	apunim	support
DICES-350		
Race=AfricanAmerican	-0.043	9077
Race=Asian	0.066	8138
DICES-990		
Age=GenX+	-0.046	14384
Age=GenZ	0.063	13741
Edu=High school and below	-0.075	7419
Gender=Man	0.028	32494
Gender=Woman	-0.025	33471
Race=AfricanAmerican	0.091	5810
Race=Latino	-0.133	6610
Race=Other	0.075	24321
Race=White	-0.120	11392
Sap		
Gender=Man	0.091	1439
Gender=Woman	-0.244	877
Kumar		
Race=AfricanAmerican	.2072	356
ToxTarget=False	−.0017	2199
ToxTarget=True	.0329	1176
Parent=No	−.0591	1868
Trans=Yes	.5322	31
ReligionImportant=No	−.1069	1069
SO=Bisexual	.1639	283
ToxProblem=Never	.2975	92

Table 2: Statistically significant ($p < 0.05$) apunim results across all four datasets. Gray rows show negative values. Consult App. E for the full results.

tators also showed the highest degree of polarization in the Kumar dataset (.532).

Finding 3. *Annotator ethnicity and gender are consistent causes of polarization.*

Group distance Figure 6 shows *apunim* trends across ordinal sociodemographic attributes listed in Table 2. We find that *polarization attribution changes consistently* as we move from one end of an attribute scale to the other–e.g., from irreligious to very religious annotators. In some sociodemographic dimensions (opinions on toxicity and religion) this is expressed in a monotonic pattern (black lines), suggesting that perceived differences become stronger between groups that are further apart ideologically or demographically. In other cases (opinions on technology and educational level), polarization attribution is caused by a few dominant modes (yellow lines). Finally, age seems to have a different effect depending on the dataset.

Finding 4. *Polarization attribution tends to grow stronger as the differences between annotator groups increase.*

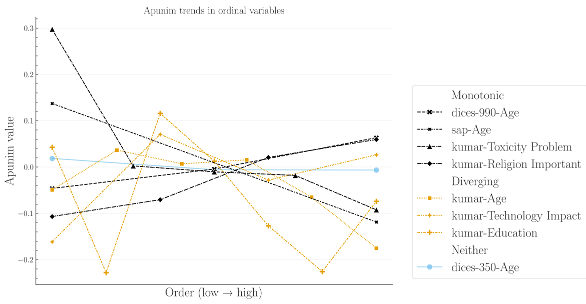

Figure 6: Polarization attribution trends in ordinal groups. The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. E.

6 Conclusion & Future work

We investigated whether polarization can be attributed to annotator groups. We discovered as of yet unknown issues regarding the presence of inherent and conflicting polarization, which we treat using a new metric, *apunim*. By applying this metric to four subjective NLP datasets, we discover that (1) race and gender seem to generally drive polarization attribution, and (2) groups that are further apart increasingly drive polarization. Finally, we estimated the robustness of our metric w.r.t. the number of annotators, and released a PyPi installable library implementing it.

While we study one group at a time, we can instead turn to iteratively exploring polarized subgroups in future work, allowing us to gradually piece together a multidimensional explanation of polarization (see Footnote 6, §3.1). It would also be interesting to explore whether LLM-as-a-judge systems reproduce the polarization patterns observed among different minority groups–which would solve the issues with annotation counts presented in our study.

7 Limitations

Researchers often release only aggregated single-label annotations or omit annotator sociodemographic characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values.

Finally, although we compare findings across datasets and relate them to prior work, we do not directly compare polarization and agreement metrics, since they rely on fundamentally different assumptions and mechanisms (see §2.2). These differences also make it difficult to fully explain certain patterns in our results, such as why some groups show statistically significant effects while others do not.

8 Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research. No identifiable, personal information is included in our pipeline. We did not perform any annotation tasks ourselves. Additionally, this study uses dichotomies across the two predominant genders, and religious groups as examples. The authors do not claim that these examples necessarily reflect the views or behaviors of these groups; they exist only for the purpose of demonstration. The controversial statements expressed in this publication also do not express the views of the authors.

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A Acronyms

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

B Annotation details

The number of annotators per comment for all datasets presented in §3.2 can be found in Figure 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 5 since they are very likely a data error.

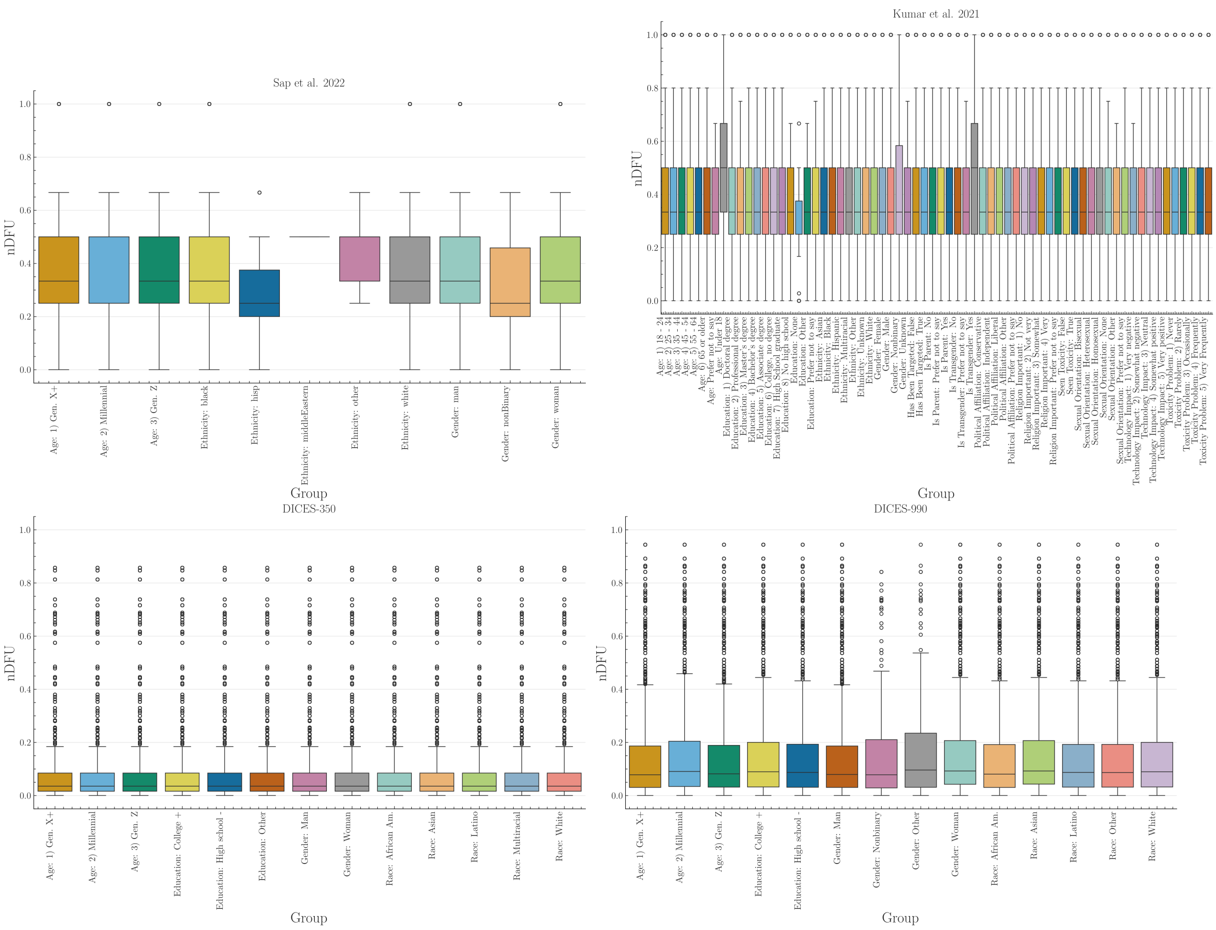

Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per sociodemographic dimension.

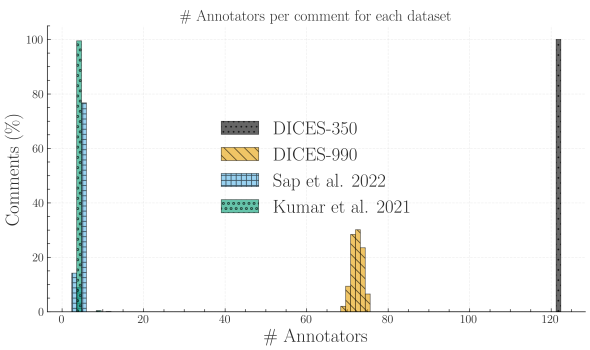

Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

C Statistical considerations

C.1 Global vs. local statistics

In §3.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset–in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

We note that when calculating apriori polarization, instead of using randomly sized splits, we create random stratified splits (i.e., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we create random partitions of sizes 80 and 20).

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

C.2 Filtering ineligible comments

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude comments annotated exclusively by a single group (i.e., comments where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per comment. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

C.3 P-value estimation

Since our test is parametric (§4.3), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per sociodemographic group to estimate nDFUs in Equations 6-8, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3.2). We discuss condition (2) in §4.4. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values–specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (`https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

D.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple sociodemographic dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$), which are originally calculated to obtain the apriori polarization values (see Equation 6), for the p-value calculations (§4.3). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Equation 6), and the number of apunim values used in the p-value calculation have to be the same.

D.3 Replication Details

We use version `1.0.2` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 6) and the p-value estimation (§4.3), and set a FWER rate of .95.

While the library efficiently handles a large number of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3.2 complete within a minute, the Kumar dataset necessitates approximately one hour of computation, even when using a 60% sample. Furthermore, the annotator variance analysis detailed in §4.4 required over ten hours. While this latter time investment is anticipated due to the inherently demanding nature and non-standard application of our library, it highlights a scaling bottleneck under specific data structures. We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each sociodemographic dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each sociodemographic subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package (Tange, 2025) for our own experiments).

In Figure 5, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

E Full Results

Group	apunim	support
Sap		
Age=GenX+	.0660	271
Age=Millennial	−.0068	1886
Ethnicity=white	.0004	1896
Gender=Man	.0907***	1439
Gender=Woman	−.2442***	877

Table 4: Apunim results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

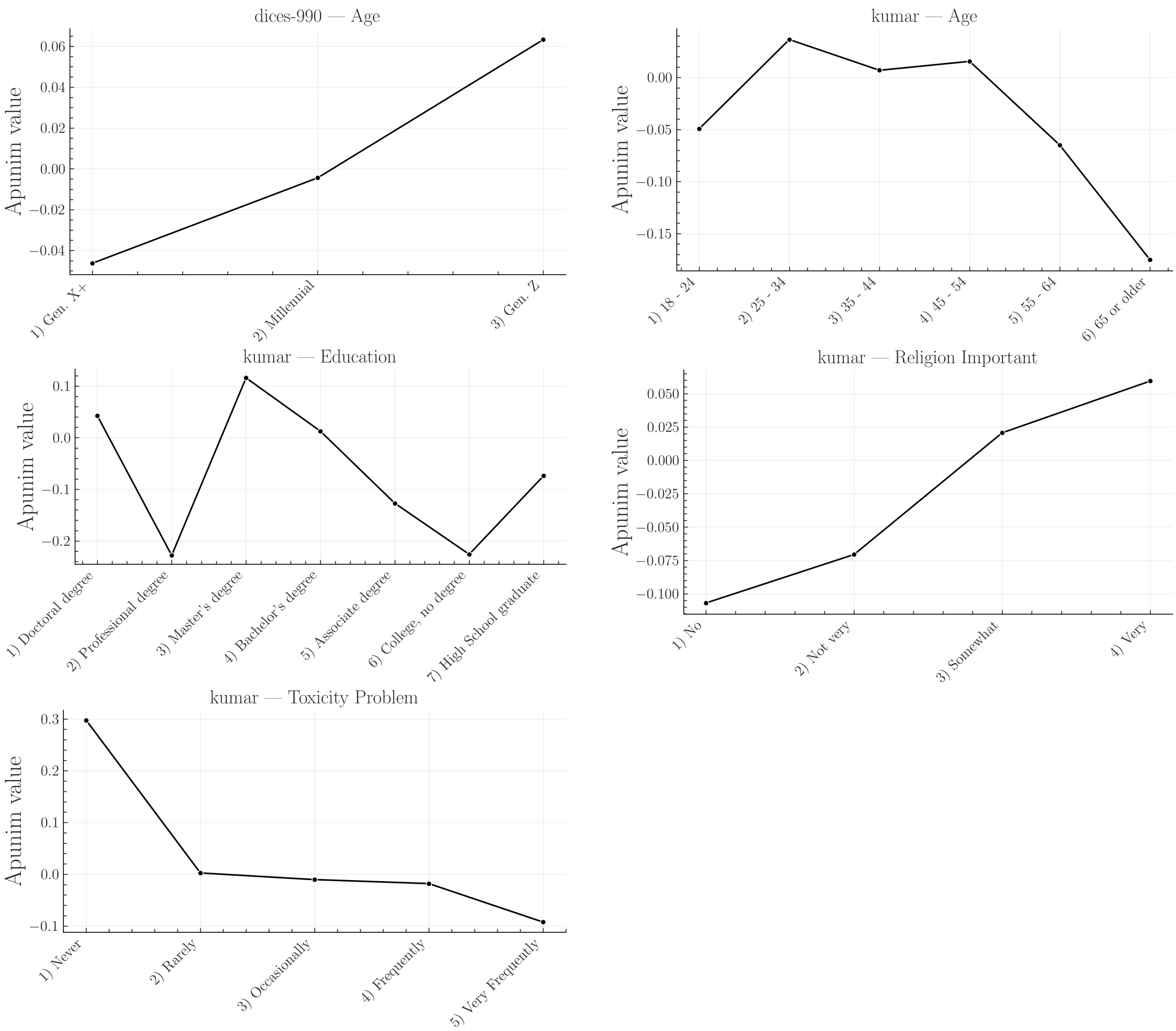

Figure 9: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal sociodemographic dimension. This figure is complementary to Figure 6. The values in the graphs were extracted from the Tables of E.

Group	apunim	support
Kumar		
Age=18-24	−.0494	223
Age=25-34	.0365	1188
Age=35-44	.0069	762
Age=45-54	.0155	291
Age=55-64	−.0650	130
Age=65+	−.1752	17
Edu=Doctoral degree	.0426	8
Edu=Professional degree	−.2278	7
Edu=Master's degree	.1159*	459
Edu=Bachelor's degree	.0126	1193
Edu=Associate degree	−.1270	176
Edu=College, no degree	−.2258***	540
Edu=High School grad	−.0739	122
Asian	.0729	75
Black	.2072***	356
Hispanic	.0255	25
Multiracial	−.1099	44
Other	.1992	17
Unknown	−.0246	7
White	−.0436	1253
Gender=Female	−.0370	2070
Gender=Male	.0422	1974
Target=False	−.0017	2199
Target=True	.0329	1176
Parent=No	−.0591***	1868
Parent=Prefer not to say	−.1858	4
Parent=Yes	.0205	2156
Trans=No	−.0143	561
Trans=Prefer not to say	.3697	7
Trans=Yes	.5322**	31
Conservative	.0552	1021
Independent	−.0610	895
Liberal	−.0419	1449
Other	−.1127	21
Prefer not to say	−.0148	54
Rel=No	−.1069***	1069
Rel=Not very	−.0706	241
Somewhat	.0206	837
Very	.0595	1199
Prefer not to say	−.2144	13
False	.0594	849
True	−.0387	2051
SO=Bisexual	.1639**	283
SO=Heterosexual	−.0353	1170
SO=Homosexual	−.0103	32
SO=Prefer not to say	−.2791	23
Tech=Very negative	.2157	4
Tech=Negative	−.1614	176
Tech=Neutral	.0705	320
Tech=Positive	−.0283	1606
Tech=Very Positive	.0263	1000
ToxProblem=Never	.2975**	92
ToxProblem=Rarely	.0024	498
ToxProblem=Occasionally	−.0105	1096
ToxProblem=Frequently	−.0181	958
ToxProblem=Very Freq	−.0925	252

Table 5: Apunim results for the dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value.

Group	apunim	support
DICES-350		
Race=AfricanAm.	−.0428***	9077
Race=Asian	.0659***	8138
Age=GenX+	.0186	9703
Age=Millennial	−.0059	11268
Age=GenZ	−.0066	17528
Edu=College or higher	.0124	23475
Edu=H. school or lower	−.0152	12833
Edu=Other	−.0214*	2191
Gender=Man	−.0193	19093
Gender=Woman	.0215	19406
Race=Latino	−.0030	6886
Race=Multiracial	−.0001	5008
Race=White	−.0198	9390

Table 6: Apunim results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	support
DICES-990		
Age=GenX+	.0186	9703
Age=Millennial	−.0044	38436
Age=GenZ	.0633***	13741
Edu=College or higher	.0094	59142
Edu=H. school or lower	−.0752***	7419
Edu=Other	−.0214*	2191
Gender=Man	.0275***	32494
Gender=Woman	−.0245***	33471
Race=AfricanAm.	.0911***	5810
Race=Asian	.0107	18428
Race=Latino	−.1331***	6610
Race=Multiracial	−.0001	5008
Race=White	−.1197***	11392

Table 7: Apunim results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.